Saykat Das

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Key competencies |

- CAR-T target biology & safety (antigen loss, fratricide, cytokine release)
- Biomarker discovery & immune profiling (flow cytometry, 10x scRNA-seq, BCR)
- CRISPR-Cas9 KI/KO, plasmid & viral vector engineering
- Recombinant protein & antibody production
- Assay development (ELISA, SPR, BLI, ELISpot)
- SOPs, documentation, cross-functional collaboration

Professional experience |

Postdoc, Oslo Cancer Cluster **Nov25-**

- Designed safety-enhanced CAR-T constructs targeting AML and solid tumors (CD37/CD33 Combi-CAR).
- Engineered retroviral plasmids for CAR expression and expansion.
- Led functional characterization using BLI killing assays, cytokine-release assays, and multicolor flow cytometry.
- Evaluated immune-escape mechanisms including antigen loss and fratricide.

Researcher, Oslo University Hospital **Feb24 – Nov25**

- Specialized in multicolor flow cytometry for cell sorting and immune phenotyping.
- Performed scRNA-seq (10x Genomics) and integrated R-based data analysis for high-dimensional datasets.
- Designed and optimized immunological and molecular biology workflows for infectious disease research, including ELISA, SPR, PCR/qPCR, ELISpot
- Developed and validated protocols, ensuring reproducibility and supporting technology transfer to collaborators.

Researcher, University of Oslo **Sep23 - Jan24**

- Performed immunorepertoire analysis for therapeutic antibody identification and characterization.
- Designed recombinant protein and antibody expression systems in bacterial and mammalian cell lines.
- Optimized aseptic mammalian cell culture workflows for in vitro functional assays.
- Prepared validation reports and documentation in line with experimental and regulatory needs.

PhD Researcher, University of Oslo **Apr20 - Sep23**

- Worked with fluorescently level cell sorting for antigen-specific cell sorting.
- Produced monoclonal antibodies from sorted B cells, characterizing antigen-antibody binding kinetics and specificity using ELISA, SPR, HPLC and ammonium release assays.

- Optimized phage-display protocol for identification of broadly neutralizing antibodies.
- Worked with vitro T cell-B cell collaboration assay to study B cell function in autoimmune disease setup.

Research Assistant, University of Oslo **Jan18 - Mar20**

- Applied molecular biology techniques (RT-qPCR, western blotting, flow cytometry) in GMP-adjacent environments.
- Developed immunohistochemistry protocols for tissue protein localization studies.
- Established in vitro T-cell–B-cell co-culture assays for functional analysis of immune interactions.
- Performed Fluorescent Immunohistochemistry for tissue resident B and T cell analysis.

Education |

Apr20–Sep23	**PhD in immunology |** Faculty of medicine, University of Oslo.
Aug17 -Sep19	**MSc in molecular biology |** Department of biosciences, University of Oslo.
Jan12 – Jul16	**BSc in biochemistry and biotechnology |** USTC, Chittagong, Bangladesh.
Jan08 - Jan11	**Diploma in medical technology |** Institute of health technology, Bangladesh.

Certificates and scholarships |

2019	**Animal care and animal fieldwork certificate (FELASA-C).**
2023-2025	**Good manufacturing practice training (verification code: 1XUk53wTCi).**
Apr19 – Oct19	**Funding from UiO: Life Science for summer project: Selection of naïve B cells for antibody production.**

Technical expertise |

- **Data Analysis and visualization:** R Studio, FlowJo, GraphPad Prism, ImageJ
- **Protein Modeling:** PyMol, AlphaFold
- **Documentation:** eLab journal, Biovia ScienceCloud

Language |

English (fluent), Norsk (Intermediate), Bengali (native).

Reference |

Will be provided if required.

Publications |

Das S, Stamnaes J, Kemppainen E, Hervonen K, Lundin KEA, Parmar N, Jahnsen FL, Jahnsen J, Lindfors K, Salmi T, Iversen R, Sollid LM. **Separate gut plasma cell populations produce auto-antibodies against transglutaminase 2 and transglutaminase 3 in dermatitis herpetiformis.** Adv Sci (Weinh). 2023 Sep;10(25):e2300401. doi: 10.1002/advs.202300401. PMID: 37424036; PMCID: PMC10477854.

Heggelund JE, **Das S**, Stamnaes J, Iversen R, Sollid LM. **Autoantibody binding and unique enzyme-substrate intermediate conformation of human transglutaminase 3.** Nat Commun. 2023 Oct 5;14(1):6216. doi: 10.1038/s41467-023-42004-z. PMID: 37798283; PMCID: PMC10556103.

Das S, Stamnaes J, Høydahl LS, Skagen C, Lundin KEA, Jahnsen J, Sollid LM, Iversen R. **Selective activation of naïve B cells with unique epitope specificity shapes autoantibody formation in celiac disease.** J Autoimmun. 2024 May 15;146:103241. doi: 10.1016/j.jaut.2024.103241. Epub ahead of print. PMID: 38754235.

Iversen R, Heggelund JE, **Das S**, Høydahl LS, Sollid LM. **Enzyme-activating B-cell receptors boost antigen presentation to pathogenic T cells in gluten-sensitive autoimmunity.** Nat Commun. 2025 Mar 10;16(1):2387. doi: 10.1038/s41467-025-57564-5. PMID: 40064932; PMCID: PMC11894174.

Saykat Das, Hans Christian Dalsbotten Aass, Julia Clare Macleod, Iselin Edlund, Dag Henrik Reikvam. **Phenotypic and genotypic alteration of HBsAg-specific B cells modulate the surface antigen loss after NUC-Stop therapy.** Manuscript in preparation.